

CAPSTONE PROJECT PROPOSAL

RESOLVIA

Investment Banking Exception Intelligence & Resolution

From Break to Verified Resolution

Capstone 2 Project Proposal

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Executive Summary

RESOLVIA is an evidence-grounded exception-intelligence and resolution-assistance solution for investment-banking post-trade operations. It begins after an existing reconciliation system identifies a break; it does not replace the matching or reconciliation engine. It investigates the exception across structured records and unstructured evidence, determines likely root cause, prepares an evidence-backed resolution proposal, routes consequential actions through human approval, and validates closure.

The project is structured around three post-reconciliation maturity levels- manual break investigation, deterministic automation/RPA for repeatable exceptions, and agentic AI for heterogeneous evidence-heavy exceptions. The core principle is- AI investigates. Humans authorize. Systems execute. AI validates.

Industry evidence shows why this is a relevant business problem. DTCC reports 98.4% same-day equity matching among its clients in Q1 2026, while SS&C GlobeOp reports 800,000 trade breaks reviewed per month, 3,680 hours returned monthly and a 20% increase in day-zero break resolution. CRISIL reports cases with 65% independent break resolution and approximately 80 onshore hours saved per week.

Problem Statement

Matching is not resolution

A reconciliation engine answers- “Does A = B?” If records disagree, it creates a break. RESOLVIA begins there. The operational question becomes- “Why did it break, what should be done, who must authorize it, and did the corrective action actually fix the exception?”

Level	Question	Technology	RESOLVIA position
0- Matching / Reconciliation	Does A = B?	Matching engines, tolerances, STP	Upstream dependency; not rebuilt
1- Manual Break Resolution	What happened and how do I fix it?	Analyst investigation across systems/docs	Controlled manual-style baseline
2- Rule/RPA Resolution	Can a known pattern be handled deterministically?	Rules, RPA, workflow automation	Conventional automation baseline
3- Agentic Resolution	Can heterogeneous evidence be investigated dynamically?	ML + NLP + RAG + LLM agents	RESOLVIA

Capco explicitly separates reconciliation matching/rule automation from break remediation and describes RPA executing corrective actions. SS&C GlobeOp provides direct evidence of digital workers investigating an existing trade-break queue.

Objective

- Develop an end-to-end workflow that converts a reconciliation break into an evidence-backed, controlled and validated resolution.
- Compare a controlled manual-style baseline, a deterministic rules/RPA-style baseline and RESOLVIA.
- Measure whether evidence-grounded agents improve investigation coverage, root-cause quality and resolution recommendations beyond deterministic rules.
- Maintain explicit human authorization for material financial, client-impacting or regulatory actions.
- Quantify benefits through investigation cycle time, effort reduction, straight-through investigation and recommendation quality.

Industry Evidence & Benchmarking

Level	Company / Institution	Year	Evidence	Implication
Matching	DTCC	2026	98.4% same-day equity matching; 2.6M CTM transactions daily	Matching is already highly automated; do not rebuild it.
Exception management	DTCC	Best-practice	>90% block exceptions fixed within 1 hour; <1% block touch rate	Exception speed is an established KPI.
RPA break resolution	SS&C GlobeOp / SS&C Blue Prism	Current case	800K trade breaks/month; 3,680 hours/month returned; +20% day-zero resolution	Large-scale post-break automation is proven.
Break resolution	CRISIL Integral IQ	2025	65% of breaks resolved independently offshore	Repeatable exceptions can be industrialised.
Reconciliation automation	CRISIL Integral IQ	2025	90%+ break-resolution accuracy; 50% time reduction; ~55% cost savings	Conventional automation can create material value.
Investigation workload	CRISIL Integral IQ	2025	99% TAT reduction; ~80 onshore hours saved/week; included break investigation and trade-document scrutiny	Investigation work can create significant capacity gains.
Maturity reality	Broadridge / FCM survey	2025/26	16% manual; 53% partial; 31% mostly automated; 0% fully automated trade-break handling	Automation maturity is uneven; full automation is not universal.
Agentic	IVP	Current	Rules engine + AI agents + human oversight; agents investigate and clear breaks	Commercial market is moving toward hybrid agentic models.
Agentic	Zensar / AWS Marketplace	Current	Multi-agent matching, break context, RCA, corrective-action recommendation and human routing	Industry is already considering Agentic Evolution in this same Space.

Scope

Base asset class- cash equities. Four core break families:

- Trade / Confirmation Break- did both parties agree on what was traded?
- Settlement Break- did the trade settle correctly?
- Position Break- do holdings agree with the external record?
- Cash Break- does expected cash movement agree with actual cash movement?

One specialist corporate-action scenario may demonstrate OCR, RAG and deadline reasoning. Derivatives, FX, securities lending, collateral/margin, a full reconciliation engine and real payment integration are excluded from v1.

Data Sources

- Synthetic structured operational data- breaks, trades, positions, cash transactions and settlements.
- Synthetic confirmations, settlement notices, custodian statements, broker/counterparty correspondence and selected scanned documents.
- Historical resolved cases containing root cause, evidence, resolution and outcome for RAG.
- Public reference data where appropriate for realistic identifiers/prices/corporate-action context.
- Published industry cases as external benchmarks, not training labels.

Methodology

1. Integrate linked structured records and evidence with source identifiers.
2. Standardise dates, identifiers, quantities, amounts and categorical fields; derive break features.
3. Analyse break distribution, severity, ageing, exposure, recurrence and temporal patterns.
4. Compare Logistic Regression, Random Forest and XGBoost for four break families using macro-F1 and class-level metrics.
5. Use OCR for selected scans and RAG for historical cases and domain evidence.
6. Orchestrate Triage, Investigation, RCA, Resolution Proposal and Validation agents.
7. Require human approval for material financial/client/regulatory actions.
8. Validate expected versus actual post-action state; CLOSE only after successful validation, otherwise REOPEN/ESCALATE.
9. Compare manual-style, rules/RPA-style and agentic workflows using operational, intelligence and control KPIs.

Agent Architecture

Agent	Purpose	Authority
Triage	Validate, classify, prioritise and route	No financial execution
Investigation	Gather and connect records, documents, emails and precedents	Read/retrieve only
RCA	Determine expected vs actual state, cause, alternatives and confidence	Recommendation
Resolution Proposal	Prepare corrective action, rationale, impact and approval need	Proposal only
Validation	Verify post-action state; CLOSE / REOPEN	No unilateral material closure

Evaluation Framework

Dimension	Manual-style	Rules/RPA	RESOLVIA
Investigation effort	Controlled process proxy	Automated path time	Agent-assisted time
Coverage	All controlled cases	Known-rule cases	Eligible cases + escalation
RCA quality	Scenario ground truth	Rule mapping	Evidence-grounded RCA accuracy
Recommendation	Ground truth	Rule outcome	Recommendation accuracy
Evidence	Human-defined	Predefined sources	RAG relevance/completeness
False closure / reopen	Tracked	Tracked	Tracked
Control	Human	Rule-governed	Human approval; unauthorized action = 0
Auditability	Case record	Workflow log	Evidence → recommendation → approval → action → validation

External industry figures are benchmarks and business justification, not targets that RESOLVIA must beat. The project will report its own controlled results and clearly separate synthetic experimental evidence from production case studies.

Expected Outcomes

- Working RESOLVIA prototype- Break → Triage → Investigation → RCA → Resolution Proposal → Human Control → Action → Validation → Close/Reopen.
- Controlled synthetic cash-equity dataset with linked structured and unstructured evidence.
- Validated ML classifier for four core break families.
- OCR + Hugging Face + RAG evidence pipeline.
- Five bounded agents with explicit authority boundaries.
- Comparative evaluation of manual-style, rules/RPA-style and agentic workflows.
- Management view of efficiency, resolution, ageing, recurrence and control KPIs.
- Business case translating measured time savings into analyst capacity and potential cost impact.

Tools & Technologies

- Python; Pandas / NumPy / scikit-learn / XGBoost
- Hugging Face Transformers / Sentence Transformers
- OCR; LLM + RAG / vector database
- SQL; Streamlit; Git/GitHub

Risks & Challenges

- Synthetic data realism
- Hallucination and evidence grounding
- False closure and reopen logic
- Over-automation and human-authorisation requirements
- Model error and class imbalance
- OCR/document quality
- Historical precedent bias
- Benchmark interpretation and avoiding unsupported universal claims

Conclusion

RESOLVIA is not a new reconciliation engine. It is a controlled investigation and resolution layer that begins where existing reconciliation technology ends- at the break. It tests whether agentic reasoning can absorb the investigative workload that remains after matching, while preserving human accountability for consequential decisions and requiring machine-verifiable evidence before closure.

Selected Published Sources

- [DTCC- Accelerated Settlement FAQs & Resources \(Q1 2026\)](#)
- [DTCC- Institutional Trade Processing Best Practices](#)
- [SS&C Blue Prism- Post-Trade Automation / SS&C GlobeOp case](#)
- [SS&C Blue Prism- SS&C GlobeOp trade reconciliation case](#)
- [CRISIL- Smart shadow accounting- 65% autonomous break resolution](#)
- [CRISIL- Simplified reconciliation- 55% cost savings](#)
- [CRISIL- 99% TAT reduction / ~80 hours saved per week](#)
- [CRISIL- Asset Servicing & Operations](#)
- [Broadridge- State of the Market- FCM Front Office](#)
- [Capco- Reconciliations- Five Trends Shaping the Future Landscape](#)
- [IVP- Reconciliation Solution](#)
- [AWS Marketplace / Zensar- Intelligent Trade Reconciliation](#)